SWYNEX TECHNOLOGIES

TASK 2: EXPLORATORY DATA ANALYSIS

Medical Appointment No-Show Dataset

Project objective :

To explore the cleaned medical appointment dataset, summarize its statistical characteristics, identify trends and patterns, investigate anomalies, and communicate evidence-based insights about appointment attendance.

Dataset	Cleaned_Dataset_Task_01.csv
Records	110,526 appointment records
Fields	14 columns
Analysis tools	Python, Pandas, NumPy, Matplotlib and Seaborn
Target field	no-show (No = attended; Yes = did not attend)

1. Dataset overview :

The dataset contains 110,526 appointment records and 14 columns. Fields include patient and appointment identifiers, gender, scheduled and appointment dates, age, neighbourhood, scholarship status, selected medical-condition indicators, SMS notification status and the no-show outcome.

The outcome field is interpreted as follows: **No** means the patient attended; **Yes** means the patient did not attend. The dataset covers appointment dates from 29 April 2016 to 8 June 2016.

2. Statistical summary :

Measure	Result	Interpretation
Total records	110,526	Appointment rows analysed
Attended	88,207 (79.81%)	Most recorded appointments were attended
Did not attend	22,319 (20.19%)	Approximately one in five appointments was missed
Mean age	37.09 years	Average patient age in the records
Median age	37 years	Half of recorded ages were 37 or below
Age range	0–115 years	Includes infants and a small number of very old ages
Mean waiting time	10.18 days	Average calendar-day gap between scheduling and appointment.
Median waiting time	4 days	Half of appointments had a waiting gap of four days or les

The mean age is 37.09 years and the median is 37 years. The interquartile age range is 18 to 55 years. The age distribution therefore spans a wide range, with substantial representation across adult age groups.

3. Trends identified :

Appointment volume over time : The appointment data is from April, May and June 2016. The monthly counts were April, 3,235; May, 80,841; June, 26,450. The volume of this data set is highest in May. This is not to be taken as indicative of a long-term seasonal trend as the data covers only a portion of three months.

Weekday Volume : Wednesday (25,867) and Tuesday (25,640) had the highest appointment counts, and Saturday only had 39 records. Weekday no-show rates ranged from 19.35% on Thursday to 21.23% on Friday. The Saturday rate is based on a very small number of records and should not be generalized.

Waiting-time pattern : The average waiting time was longer among missed appointments (15.83 days) than attended appointments (8.75 days). The median was 11 days for missed appointments and 2 days for attended appointments. This is an association in the observed data, not proof that waiting longer causes a patient to miss an appointment.

4. Patterns discovered :

There were 71,839 records for females and 38,687 records for males. The no-show rates were around 20.31% of female records and 19.97% of male records. The difference is negligible.

Age groups : The age bands with the highest no-show rate were teenagers (13–18 years) at approximately 26.05% and young adults (19–30 years) at 24.71%. Seniors (61 to 75) had a lower rate of around 14.86 percent. These bands are analytical categories selected for this project.

SMS : The no show rate was 27.57% for records with an SMS and 16.70% for records without an SMS. This is not evidence that SMS reminders cause missed appointments.

Medical-condition indicators : Hypertension was mentioned in 21,801 cases, diabetes in 7,943, alcoholism in 3,360 and disability in 2,459. These are counts of appointment records where the indicator is set, not necessarily unique patients.

Neighborhood : The largest number of appointments was registered in Jardim Camburi (7,717), followed by Maria Ortiz (5,805) and Resistência (4,431). Neighborhood no-show rates should be read in conjunction with the number of records in each area; very small groups can produce unstable percentages.

5. Anomalies and data-quality observations :

Check	Finding	Interpretation / action
Missing values	0 across the loaded columns	No null values detected in this file.
Exact duplicate rows	0	No exact duplicate records detected.
Negative ages	0	No negative patient ages found.
Age above 100	7 records	Review these values; they may be valid but are unusual.
Negative waiting days	5 records	Appointment date precedes scheduled date; inspect as date/data anomalies
Long waiting-time outliers	5,803 records above 37.5 days	IQR upper fence; these are statistical outliers, not automatically erros.

6. Five key insights :

1. Overall attendance: 79.81% of appointments were attended and 20.19% were missed. The missed-appointment share is substantial and is a key measure for service planning.

2. Waiting time is associated with missed appointments: Missed appointments had a median waiting time of 11 days compared with 2 days for attended appointments. This suggests that longer waits may be a useful factor to investigate further.

3. No-show rates differ by age band: Teenagers had the highest rate among the age groups used (26.05%), while the 61–75 group had a lower rate (14.86%). Age bands may help target further investigation and communication strategies.

4. SMS status shows a notable difference: The no-show rate was 27.57% for records marked as having received an SMS versus 16.70% for records without an SMS. This is an observational association and may reflect how reminders were allocated; it is not evidence that SMS causes non-attendance.

5. Data-quality checks found specific records to review: Five records had negative waiting periods and seven had ages above 100. In addition, 5,803 waiting periods exceeded the IQR outlier threshold. These cases warrant validation before any data removal or correction.

7. Questions addressed by this EDA :

Task question	What this analysis provides
What are the important statistics?	Record counts, attendance percentages, age summary and waiting-time summary.
What trends exist?	Monthly appointment volumes, weekday volumes and weekday no-show rates. and neighbourhood.
What patterns exist?	Attendance differences by gender, age group, SMS status, medical indicators
Are there anomalies?	Checks for missing values, duplicates, unusual ages, negative waiting times and IQR outliers
What are at least five useful insights?	Five evidence-based findings are listed in Section 6.

8. Conclusion :

The exploratory analysis indicates that most appointments were attended, but roughly one in five was missed. Waiting time, age group and SMS status show differences in observed no-show rates, while weekday and monthly appointment volumes vary across the limited date range.

Data checks found no missing values or exact duplicate rows, but identified a small number of unusual age and date records and a larger group of long-wait outliers for review.

These findings are descriptive. They identify relationships and areas for further investigation, but do not establish causation. Before making operational recommendations, the results should be validated with additional time periods and relevant context.

9. Methodology and limitations :

The analysis used Python with Pandas, NumPy, Matplotlib and Seaborn. Categorical attendance rates were calculated as the percentage of records with no-show = Yes. Waiting time was calculated as the difference in calendar dates between appointment day and scheduled day. The IQR method flagged waiting-time values above Q3 + 1.5 × IQR.

Limitations: the data covers a short period (29 April–8 June 2016); appointment rows are not necessarily unique patients; neighbourhood percentages can be unstable for small groups; and observed associations (especially SMS status) should not be interpreted as causal effects. Age above 100 and long waiting times require source validation rather than automatic deletion.